Nama : Meitha Noer Meidayanti
NIM : 20134410019
Kelas : 2NK

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Soal Latihan

1) Sebuah material dielektrik dikenai medan listrik dengan persamaan : $\vec{E} = 0{,}1 \cos(2\pi \times 10^6 t)\, \vec{a}_x$ V/m, material tersebut adalah polystyrene dengan $\varepsilon_{r_1} = 2{,}5$ dan distilled water $\varepsilon_r = 81$

Tentukan :

a) Vektor polarisasi
b) Suseptibilitas elektrik
c) Rapat arus polarisasi
d) Rapat muatan polarisasi

Pada kedua material dielektrik tersebut

Jawab

$\vec{E} = 0{,}1 \cos(2\pi \times 10^6 t)\, \vec{a}_x$ V/m

$\varepsilon_{r_1} = 2{,}5$

$\varepsilon_{r_2} = 81$

a) $\vec{P}_1 = \varepsilon_0 \cdot \chi_{e_1} \vec{E}$

$= 8{,}854 \cdot 10^{-12} \cdot (2{,}5 - 1)\ 0{,}1 \cos(2\pi \cdot 10^6 t)\, \vec{a}_x$

$= 1{,}3281 \cdot 10^{-12} \cos(2\pi\, 10^6 t)\, \vec{a}_x$

$\vec{P}_2 = \varepsilon_0 \chi_{e_2} \vec{E}$

$= 8{,}854 \cdot 10^{-12} \cdot (81 - 1) \cdot 0{,}1 \cos(2\pi \cdot 10^6 t)\, \vec{a}_x$

$= 70{,}832 \cdot 10^{-12} \cos(2\pi \cdot 10^6 t)\, \vec{a}_x$

b) $\chi_1 = \varepsilon_{r_1} - 1$ $\qquad \chi_2 = \varepsilon_{r_2} - 1$

$= 2{,}5 - 1$ $\qquad = 81 - 1$

$= 1{,}5$ $\qquad = 80$

c) $\vec{J}_{P_1} = \dfrac{\partial \vec{P}}{\partial t}$

$= \dfrac{\partial (1{,}3281 \cdot 10^{-12} \cos(2\pi \cdot 10^6 t)\, \vec{a}_x)}{\partial t}$

$= 1{,}3281 \cdot 10^{-12} \cdot 2\pi \cdot 10^6 (-\sin[2\pi \cdot 10^6 t])\, \vec{a}_x$

$= -8{,}344 \cdot 10^{-6} \sin(2\pi \cdot 10^6 t)\, \vec{a}_x$

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$$\vec{J}_{P_2} = \frac{\partial \vec{P}_2}{\partial t}$$

$$= \frac{\partial \left(70,832 \cdot 10^{-12} \cos (2\pi \cdot 10^6 t)\, \vec{a}_x \right)}{\partial t}$$

$$= 70,832 \cdot 10^{-12} \cdot 2\pi \cdot 10^6 \left(-\sin (2\pi \cdot 10^6 t)\, \vec{a}_x\right)$$

$$= -445,05 \cdot 10^{-6} \sin (2\pi \cdot 10^6 t)\, \vec{a}_x$$

d) $\rho_{P_1} = -\nabla \cdot \vec{P}_1$

$\vec{P}_1 = 1,3281 \cdot 10^{-12} \cos (2\pi \cdot 10^6 t)\vec{a}_x$

$P_{1x} = 1,3281 \cdot 10^{-12} \cos (2\pi\, 10^6 t)$

$P_{1y} = 0$

$P_{1z} = 0$

$$-\nabla \cdot \vec{P}_1 = -\left(\frac{\partial (1,3281 \cdot 10^{-12} \cos (2\pi \cdot 10^6 t))}{\partial x} + \frac{\partial (0)}{\partial y} + \frac{\partial (0)}{\partial z} \right)$$

$\rho_{P1} = 0$

$\rho_{P2} = -\nabla \cdot \vec{P}_2$

$\vec{P}_2 = 70,832 \cdot 10^{-12} \cos (2\pi\, 10^6 t)\vec{a}_x$

$P_{2x} = 70,832 \cdot 10^{-12} \cos (2\pi \cdot 10^6 t)$

$P_{2y} = 0$

$P_{2z} = 0$

$$-\nabla \cdot \vec{P}_2 = -\left(\frac{\partial (70,832 \cdot 10^{-12} \cos (2\pi \cdot 10^6 t))}{\partial x} + \frac{\partial (0)}{\partial y} + \frac{\partial (0)}{\partial z} \right)$$

$\rho_{P2} = 0$

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2. Dua buah konduktor masing-masing memiliki konduktivitas berturut-turut $\sigma_1 = 5,7 \cdot 10^7$ (S/m) dan $\sigma_2 = 3,2 \cdot 10^7$ (S/m). Jika kedua konduktor ini dikenai medan listrik $\vec{E} = 10\vec{a}_y$ (V/m) tentukan perbandingan rapat arus konduksi konduktor 1 terhadap 2.

Jawab

$\sigma_1 = 5,7 \cdot 10^7$ (S/m)

$\sigma_2 = 3,2 \cdot 10^7$ (S/m)

$\vec{E} = 10\vec{a}_y$ (V/m)

$\frac{\vec{J}_{c1}}{J_{c2}} = \ldots ?$

$\frac{\vec{J}_{c1}}{J_{c2}} = \frac{\sigma_1 \cdot \vec{E}}{\sigma_2 \vec{E}}$

$= \frac{5,7 \cdot 10^7}{3,2 \cdot 10^7}$

$= \frac{57}{32}$